

Camille Casas, Rigoberto Padilla, Min Ma Kha, Samantha Mireles, Chris Roy

Professor Rao

ITAI 1371

09/03/2026

Reflection Journal

[Min Ma Kha]

Working through this lab helped me realize that machine learning is not just about choosing the right algorithm, but also about choosing and understanding the data you give the model. The Logistic Regression model achieved 88.9% accuracy, while the Decision Tree reached 83.3%, but testing different features changed the results. By switching the features to flavanoids, color_intensity, and proline, my model achieved 91.7% accuracy. This showed me how important feature selection can be.

The confusion matrix was also helpful because it showed exactly where the models were making mistakes instead of only giving an overall accuracy score. One thing I found interesting is that getting a higher test score does not automatically mean the model is better. Since the test data was used to choose the best features, more testing with cross-validation would be needed to confirm the results. Overall, this lab showed me that data preparation, feature selection, and evaluation are just as important as the algorithm itself.

In the feature experiment, I set random_state=84 in the train/test split, achieving an accuracy of 94.4% (34 out of 36 test wines identified correctly). What surprised me was the importance of that single number. The random_state affects which wines get into the sample for testing, meaning that changing only that parameter will not change the data, the features used or the model itself; it will only change the sample of wines that will be graded. Since the test sample happens to be small, so that contained only 36 wines, each wine accounts for 2.8% of the score and a different sample could change the result by a number of points. I also learned that the random_state for LogisticRegression does not matter as the default lbfgs solver is deterministic; therefore, all of the variations were due to the random_state parameter. Thus, I concluded that the accuracy obtained in only one train/test split was more like a snapshot than a measurement of something; accordingly, I need to use cross-validation to obtain accurate results.

[Samantha Mireles]

Working on this lab helped me better understand how the data and features we choose can affect the performance of a machine model. In my section, I learned about different types of data used in machine learning and then experimented with selecting different features from the Wine dataset.

I chose alcohol, hue and proline and tested them using Logistic Regression. I was surprised that the model reached about 97.2% accuracy with these three features. Before this lab, I did not fully understand why feature selection was important, but seeing the accuracy change made the concept much clearer to me. I also learned that building a machine learning model is not only about running code.

You have to look at the results, compare different choices, and understand why a model may perform better or worse. Overall, this lab gave me a better understanding of the machine learning workflow and helped me see how experimenting with data can improve a model.

[Rigoberto Padilla]

This lab helped me understand the basic machine learning workflow.

I learned the differences between supervised, unsupervised, and reinforcement learning.

Working with the Wine dataset helped me understand features and classes.

I also learned why preparing and exploring data is important.

Overall, I better understand how a machine learning model is built and evaluated.

[Chris Roy]

This lab helped me brush my python knowledge up, as well as helped me understand how to use the Sklearn library more efficiently, since I use it for ITAI 1378 as well. I am also able to learn how to use matplotlib more efficiently since I used to only barely use it for the Tkinter library as that was my forte in python.

[Camille Casas]

This lab helped me better understand how preprocessing, feature selection, training, and evaluation work together in machine learning. Comparing my model's 83.3% accuracy to the original 88.9% showed me that changing features does not always improve performance. The confusion matrix also helped me see where the model performed well and where it struggled. Overall, I learned that even a lower-performing experiment can provide valuable insight and help improve future models.